

Midterm N°2

Name : _____ Groupe: _____

Exercise 1. Among these assertions, specify which ones are true and which ones are false.

	True	False
1- If $P(A \cup B) = 0.5$ and $P(A \cap B) = 0.5$, then $P(A) = P(B)$.		
2- if $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$ then $P(A \cap B) = 0.3$.		
3- If $P(A B) = 0.5$ and $P(B) = 0.5$, then A and B are necessarily independent.		
4- If two events are independent, then $p(A B) = p(B A)$.		
5- If two events are independent then both $P(A B) = P(A)$ and $P(B A) = P(A)$.		
6- If $P(A B) = 0.6$ and $P(B) = 0.2$ then $P(A \cap B) = 0.12$.		
7- If $P(A \cap B) \geq 0.10$ then $P(A) \geq 0.10$.		
8- If A and B are mutually exclusive events, then $P(A B) = 0$.		
9- Bayes' rule states that $p(A B) = p(B A)$.		
10- A certainty event has a probability equal to 0.9.		
11- Any random variable is a measurable function.		
12- A random variable can be qualitatif		
13- $X \stackrel{d}{=} Y$ if $\forall x \in \mathbb{R}, F_X(x) = F_Y(x)$.		
14- $x_1 \leq x_2 \implies F(x_1) \leq F(x_2)$.		
15- B and $A \cap B^c$ are disjoint.		
16- $P(A) - P(B) \neq P(A \cap B^c)$.		
17- If $A_i =]0, \frac{1}{i}[$ then $\lim_{i \rightarrow \infty} P(A_i) = 0$.		
18- $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$.		
19- $P(A_1 \cup A_2 \cup \dots) \geq P(A_1) + P(A_2) + \dots$		
20- $E[XY] = E[X]E[Y]$.		

Exercise 2. We give the following table

Number of units (x_i)	1	2	3	4	5	6	7	8	9	10
Unit price (y_i)	1504	809	540	438	375	320	290	254	231	212

- Calculate the correlation coefficients of pearson and spearman then explain the difference between the two obtained results

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2. Set $z_i = \frac{1}{x_i}$, calculate z_i then draw the scatter plot between z_i and the unit price y_i . Comment.



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Exercise 3. Consider the data in the table below. In this table Y is the purity of oxygen produced in a chemical distillation process, and X is the percentage of hydrocarbons that are present in the main condenser of the distillation unit.

Obs. N°	X	Y						
1	0.99	90.01						
2	1.02	89.05						
3	1.15	91.43						
4	1.29	93.74						
5	1.46	96.73						
6	1.36	94.45						
7	0.87	87.59						
8	1.23	91.77						
9	1.55	99.42						
10	1.40	93.65						
11	1.19	93.54						
12	1.15	92.52						
13	0.98	90.56						
14	1.01	89.54						
15	1.11	89.85						

1. Measure the correlation between the two variables then interpret it.

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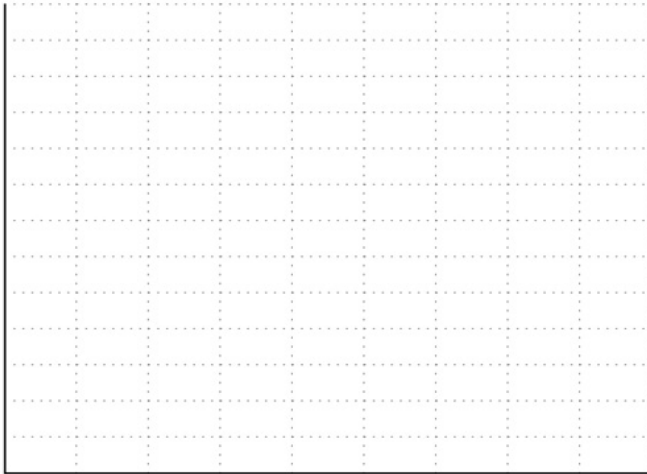
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2. Draw the scatter plot between the two variables. Which model is the most appropriate to fit our data.



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3. We aim to explain in both sens the relationship between Y and X . Let

$$Y = aX + b + \varepsilon \quad \text{and} \quad X = a'Y + b' + \varepsilon'.$$

Give the values of the parameters a, a', b and b' for which $\sum \varepsilon^2$ and $\sum (\varepsilon')^2$ are minimal.

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4. Plot in the above figure the two lines $D_{Y|X}$ and $D_{X|Y}$ represented by the two equations above. What are the coordinates of the point of intersection between the two lines and what does it represent?

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5. Compute the residual variance and the variance explained by the regression line $D_{Y|X}$. Do you think that the regression line fit well the data, measure it.

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6. According to you, what is the level of hydrocarbon required to achieve 100% oxygen purity and what is the least percentage of oxygen purity at a hydrocarbon level of 0.8.

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